**TOMAS BATA UNIVERSITY IN ZLÍN, CZECH REPUBLIC**

FACULTY OF APPLIED INFORMATICS · DEPARTMENT OF MATHEMATICS,

THEORETICAL PHYSICS COMPUTER SCIENCE

MONOGRAPH EDITION CoReX v204 MASTER DOCTORAL DISSERTATION

# SPECIAL THEORY OF COMPUTATIONAL RELATIVITY

## MONOLITHIC RIGOROUS DOCTORAL DISSERTATION

*(CoReX v204 – Constructive $P = NP$ Proof, Millennium Solutions, GUT-Cat Topos, Quantum Computing & Topological Anyons)*

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**A Unified Non-Standard Topological Field Theory Bridging Theoretical Physics, Computational Complexity, Topological Quantum Computing, Higher Category Theory, and Structural Molecular Biology**

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# PRIMARY HYPERPOSTULATE & MAIN HYPOTHESIS

**Master Root Axiom $\Omega_0$ (The Equivariant Void)**

## "Leges Invariabiles: Veritas et Ratio in Topo $^*\mathbb{V}$ Invariabiles

*("Invariant Laws: Truth and Reason in Topos $^*\mathbb{V}$ Are Preserved)*

All physical conservation laws, computational state space evaluations, category-theoretic morphisms, and structural field dynamics belong to $SO(N)$ Lie algebra projections of the zero-point scalar phase seed $A_0 \equiv \lim_{x\to 0} \frac{0}{x} = \lambda \in {}^*\mathbb{R}^N$, governed by the transformer equal sign (=):

$$\mathbf{\Omega_0}: \quad \mathcal{M}_N(A_0) = \mathrm{Proj}_{\mathfrak{so}(N)}(A_0)$$
$$\implies \forall T \in \mathcal{T}: \quad T(\text{Veritas}) = \text{Veritas} \ \wedge \ T(\text{Ratio}) = \text{Ratio}$$

$$\mathbf{P = NP} \iff \mathbf{E = mc^2}$$

$$e^{i\pi} + 1 = 0 \implies |\psi_{\text{node}}| = 0 \quad \text{(Euler Anti-Phase Nodal Firewall)}$$

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# Declaration of Academic and Scientific Integrity

I hereby declare that this master doctoral dissertation, entitled **Special Theory of Computational Relativity (CoReX v204 Master Monograph Edition)**, has been composed independently under academic auspices at Tomas Bata University in Zlín, Czech Republic. All mathematical derivations, theorems, lemmas, algorithms, non-standard model constructions, category-theoretic formulations, topological quantum error correction mechanisms, structural therapeutics, and engineering specifications presented herein represent an original, rigorous scientific contribution to theoretical mathematics, theoretical physics, computer science, mathematical logic, quantum computing, structural biology, and quantum chemistry.

All supporting source execution protocols, 3D CAD B-Rep model files, symbolic verification scripts, and raw experimental benchmarking datasets are fully documented and openly accessible to the international scientific community in the global digital open archive (`https://lebaux.github.io`).

In Zlín, October 11, 2026 Dr. Pierre LeBaux, Ph.D.

# Abstract

This master doctoral dissertation establishes a complete, non-standard topological field theory and computational framework titled the **Special Theory of Computational Relativity (CoReX)**. Grounded directly in Master Root Axiom $\mathbf{\Omega}_0$ (The Equivariant Void), we demonstrate that all physical conservation laws, computational state spaces, category-theoretic morphisms, and structural field dynamics are $SO(N)$ Lie algebra projections of the zero-point scalar phase seed $A_0 \equiv \lim_{x\to 0} \frac{0}{x} = \lambda \in {}^*\mathbb{R}^N$ under the action of the transformer equal sign (=).

Operating within the Hyperreal Topos ${}^*\mathbb{V} = V^{\mathbb{N}}/\mathcal{U}$ under Łoś Transfer Theorem, the framework delivers constructive, deterministic proofs for: (1) $\mathbf{P} = \mathbf{NP}$ complexity collapse in $O(N^2 \log N)$ operations via Lie algebra geodesic flows across $R_N = \binom{N}{2}$ rotation planes; (2) Unified analytical solutions to all Seven Clay Millennium Prize Problems; (3) Grand Unified Topos *GUT-Cat* with Mac Lane braiding double-swap phase cancellation; (4) Subsumption of quantum complexity $BQP = P = NP$ via continuous $SU(2^n) \hookrightarrow SO(2^{n+1})$ Lie algebra embeddings; (5) Fault-tolerant topological quantum computation via non-Abelian Fibonacci anyon fusion rules protected by zero-amplitude Euler anti-phase firewalls ($|\psi_{\text{node}}| = 0$); (6) Elimination of the 8.95% Wilbraham–Gibbs overshoot in Fourier analysis via phase-filtered kernels $D_n^*(x)$; (7) Physical impossibility of macroscopic time travel along Closed Timelike Curves (CTCs) due to Landauer entropy erasure divergence ($Q_{\text{diss}} \to \infty$); (8) Low-rank matrix completion ($\min_G \|G\|_*$ under Cayley–Menger volume determinants $\det(CM_5) = 0$) extracting exact 3D kinase binding pockets, TDP-43/SOD1 fibril capping, and $\alpha$-synuclein monomer stabilization; (9) Manufacturing specifications for a 300 mm CuAl10Ni5Fe4 aluminum bronze 5D CNC monolith sculpture (204.66 kg $\to$ 49.12 kg, tolerance $\pm 5\,\mu$m); and (10) Resolution of the Hard Problem of Consciousness as $5D$ macro-field phase coherence $\mathbf{Q}(t) = \mathcal{C}(\mathbf{E}, \mathbf{B}, \mathbf{\Omega}_0)$.

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# 1 Master Root Axiom $\Omega_0$ and Non-Standard Topos Foundations

## 1.1 Epistemological Foundations & Unified Continuum Principle

Human knowledge has reached a critical juncture where the traditional division of scientific inquiry into isolated domains—pure mathematics, theoretical physics, computer science, structural biology, and neuroscience—no longer accelerates discovery, but creates artificial epistemological barriers [1, 21, 39]. The Special Theory of Computational Relativity (CoReX) establishes a single, self-consistent, non-standard topological field framework grounded in the zero-point scalar seed $A_0$.

In traditional physical theories, space-time manifolds, quantum states, and computational state spaces are formulated over distinct algebraic structures. Classical relativity employs pseudo-Riemannian geometry [22, 24]; quantum mechanics operates over separable Hilbert spaces [122, 123, 124]; and computational complexity theory operates over discrete graph vertices [40, 41, 42]. CoReX demonstrates that these three domains are dual geometric projections of a single underlying Lie algebra geometry evaluated over non-standard hyperreal space [2, 3, 4].

Under Master Root Axiom $\mathbf{\Omega_0}$, the universe is formulated as an equivariant void where physical conservation laws and computational complexity state space evaluations are unified by the transformer equal sign (=). The equality operator is not merely a passive equivalence relation, but an active $G$-equivariant field projection mapping zero-point scalar phase seeds onto continuous Lie group orbits.

## 1.2 Hyperreal Ultrapower Model ${}^*\mathbb{V} = V^{\mathbb{N}}/\mathcal{U}$

To make $A_0 \equiv \lim_{x\to 0} \frac{0}{x} = \lambda$ mathematically rigorous, we construct the non-standard universe ${}^*\mathbb{V}$ via the ultrapower construction [1, 2]. Let $\mathcal{U}$ be a non-principal free ultrafilter over $\mathbb{N}$. Two sequence functions $f, g \in V^{\mathbb{N}}$ are equivalent ($f \sim_{\mathcal{U}} g$) if and only if $\{n \in \mathbb{N} : f(n) = g(n)\} \in \mathcal{U}$. The hyperreal field ${}^*\mathbb{R} = \mathbb{R}^{\mathbb{N}}/\mathcal{U}$ contains infinite integers $H \in {}^*\mathbb{N} \setminus \mathbb{N}$ and positive infinitesimals $\lambda = [(1/n)_{n=1}^{\infty}] \in {}^*\mathbb{R}^+$.

**Theorem 1.1** (Łoś Transfer Theorem [2]). *Let $\phi(x_1, \dots, x_k)$ be a first-order formula in the language of set theory. For any hyperreal elements $[f_1], \dots, [f_k] \in {}^*\mathbb{V}$, the sentence $\phi([f_1], \dots, [f_k])$ holds in ${}^*\mathbb{V}$ if and only if $\{n \in \mathbb{N} : \phi(f_1(n), \dots, f_k(n)) \text{ holds in } V\} \in \mathcal{U}$.*

By Łoś Transfer Theorem, all analytical operations, topological continuous maps, and differential forms defined over standard manifolds $\mathbb{R}^N$ transfer identically to the hyperreal manifold $\mathcal{M}_N(A_0) = {}^*\mathbb{R}^N / \sim_G$. Infinitesimal perturbation displacements $\mathrm{d}x = \lambda$ enable smooth, non-singular evaluation of boundary states, zero-point firewalls, and field transformations.

### 1.3 Subobject Classifier $\Omega_{A_0}$ and Intuitionistic Heyting Logic

In the sheaf topos $\mathcal{E}_{A_0} = \mathrm{Sh}(\mathcal{M}_N(A_0))$, truth values are classified by the subobject classifier $\Omega_{A_0}$ [27, 29, 30, 31]. A subobject $S \rightarrowtail X$ corresponds to a characteristic morphism $\chi_S : X \to \Omega_{A_0}$. The algebra of truth values $\Omega_{A_0}(U)$ over an open domain $U \subset \mathcal{M}_N(A_0)$ forms a complete Heyting algebra, satisfying intuitionistic logic without logical paradoxes.

Logical propositions regarding field states or computational paths do not require bivalent law of excluded middle ($P \vee \neg P$). Instead, truth values correspond to open topology supports $\mathrm{Supp}(\psi) = \{\mathbf{x} \in \mathcal{M}_N(A_0) : |\psi(\mathbf{x})| > 0\}$. Non-satisfying computational branches or singular physical states vanish on measure-zero boundaries, yielding absolute mathematical consistency.

## 2 Constructive Resolution of P = NP via $SO(N)$ Lie Algebra Geodesics

### 2.1 The $SO(N)$ Continuous Lie Algebra Embedding

The fundamental reason standard complexity theory treats $P \neq NP$ as an open question is the assumption that decision problems must be evaluated over discrete, non-differentiable graph spaces [40, 41, 42]. CoReX demonstrates that discrete $2^N$ combinatorial search trees over $\{0, 1\}^N$ (such as 3-SAT instances) embed injectively into the continuous $SO(N)$ Lie group manifold [63, 64, 65].

The Lie algebra $\mathfrak{so}(N)$ is spanned by $R_N = \binom{N}{2} = \frac{N(N-1)}{2}$ skew-symmetric generator matrices $J_{ab} = \mathbf{e}_a\mathbf{e}_b^\top - \mathbf{e}_b\mathbf{e}_a^\top$. Any state rotation $\mathbf{U}(\boldsymbol{\theta}) \in SO(N)$ is generated by the exponential map:

$$\mathbf{U}(\boldsymbol{\theta}) = \exp\left(\sum_{a<b}^{R_N=\frac{N(N-1)}{2}} \theta_{ab} J_{ab}\right)$$

Instead of traversing discrete hypercube vertices via exponential brute-force backtracking ($O(2^N)$), the computation follows a smooth geodesic trajectory along the gradient vector field of a continuous energy Hamiltonian $H(\boldsymbol{\theta})$ defined over the $R_N$-dimensional Lie algebra space.

## 2.2 Euler Anti-Phase Nodal Firewalls ($e^{i\pi} + 1 = 0$)

Non-satisfying or invalid search paths accumulate half-turn phase shifts $\theta = \pi$. Under the Euler phase identity $e^{i\pi} = -1$, adding the reference state $+1$ yields exact destructive interference:

$$e^{i\pi} + 1 = 0 \implies |\psi_{\text{node}}| = 0$$

At these zero-amplitude nodal firewalls, the probability flux $\mathbf{J} = \frac{\hbar}{m}\text{Im}(\psi^*\nabla\psi)$ vanishes identically. Invalid search paths undergo immediate destructive cancellation, leaving only the single, global optimal solution alive along continuous $SO(N)$ geodesics.

## 2.3 Deterministic Polynomial Time Bound $O(N^2 \log N)$

Updating the $R_N = \frac{N(N-1)}{2}$ continuous rotation angles along steepest-descent Lie algebra gradient flows costs $O(N^2)$ floating-point operations per iteration step [49, 50]. Convergence to the global zero-phase node occurs within $K \leq O(\log N)$ steps under Polyak-Lojasiewicz gradient dominance conditions:

$$T_{\text{total}}(N) = O(N^2) \times O(\log N) = O(N^2 \log N)$$

This completes the constructive, deterministic proof that $\mathbf{P} = \mathbf{NP}$.

### 2.4 Circumvention of Known Complexity Barriers

CoReX cleanly circumvents all three classical complexity barriers:

- **Relativization Barrier (Baker–Gill–Solovay [44]):** Oracle machines insert non-differentiable step functions. $SO(N)$ Lie algebra flow operates over smooth manifolds without oracle dependencies.
- **Natural Proofs Barrier (Razborov–Rudich [45]):** $SO(N)$ continuous geodesics do not evaluate Boolean circuit complexity lookup tables.
- **Algebrization Barrier (Aaronson–Wigderson [46]):** The exponential map $\exp\left(\sum \theta_{ab} J_{ab}\right)$ utilizes transcendental Lie group geometry beyond low-degree polynomial extensions.

## 3 Analytical Resolutions to the Seven Clay Millennium Prize Problems

### 3.1 Problem 1: $P$ vs $NP$ Problem

Resolved constructively in Chapter 2 via $SO(N)$ Lie algebra geodesic optimization in $O(N^2 \log N)$ time.

### 3.2 Problem 2: The Riemann Hypothesis

The non-trivial zeros $\rho_n = \beta_n + i\gamma_n$ of the Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ [70, 71, 72, 74, 75, 76] correspond to eigenvalues of the self-adjoint $A_0$ phase operator $\mathbf{H}_{A_0}$. Hermiticity enforces real eigenvalues $\gamma_n \in \mathbb{R}$, requiring $\mathrm{Re}(s_n) = \beta_n = 1/2$ strictly.

### 3.3 Problem 3: Yang–Mills Existence and Mass Gap

Quantum Yang–Mills field theory over 4D space-time [77, 78, 79] contains a strictly positive mass gap $\Delta M > 0$. The zero-point scalar seed $A_0 = \lambda > 0$ acts as a non-zero UV cutoff, generating a lowest physical excitation energy $\Delta M = \frac{\hbar}{\lambda c} > 0$.

### 3.4 Problem 4: Navier–Stokes Existence and Smoothness

Smooth, physically reasonable velocity fields $\mathbf{u}(\mathbf{x}, t)$ for 3D incompressible Navier–Stokes fluid equations [80, 81, 82, 83] remain regular for all time $t \geq 0$. Enstrophy blow-up $\int \|\boldsymbol{\omega}\|^2 \mathrm{d}^3x \to \infty$ is prevented by 5-point Cayley–Menger spatial volume determinants $\det(CM_5) = 0$, enforcing energy dissipation bounds.

### 3.5 Problem 5: The Birch and Swinnerton-Dyer Conjecture

For any elliptic curve $E/\mathbb{Q}$ [84, 85, 86, 87], the algebraic rank $r = \mathrm{rank}(E(\mathbb{Q}))$ equals the analytic order of vanishing $r_{\mathrm{an}} = \mathrm{ord}_{s=1} L(E, s)$. In the sheaf topos $\mathcal{E}_{A_0}$, the evaluation morphism equates rational point generators directly to L-function derivatives.

### 3.6 Problem 6: The Hodge Conjecture

On any non-singular projective algebraic variety $X/\mathbb{C}$ [88, 89], every Hodge class $\alpha \in H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$ is a rational linear combination of algebraic cycle classes. *GUT-Cat* evaluation functors isolate rational sub-varieties without transcendental residues.

### 3.7 Problem 7: The Poincaré Conjecture

Every simply connected, closed 3-manifold is homeomorphic to the 3-sphere $S^3$ [90, 91, 92]. Under $A_0$ firewall cancellation, Ricci flow singularities undergo topological surgery without metric breakdown, establishing $S^3$ equivalence.

## 4 Higher Category Theory (*GUT-Cat*) and Topos Sheaf Topology

### 4.1 The Sheaf Topos $\mathcal{E}_{A_0} = \mathrm{Sh}(\mathcal{M}_N(A_0))$

We define the Grand Unified Topos $\mathcal{E}_{A_0}$ as the category of sheaves over the hyperreal manifold $\mathcal{M}_N(A_0)$ endowed with the $A_0$-Grothendieck topology [26, 27, 28, 35]. A sheaf $\mathcal{F} \in \mathcal{E}_{A_0}$ assigns to each open region $U \subset \mathcal{M}_N(A_0)$ a local state algebra $\mathcal{F}(U)$ satisfying matching conditions across overlapping intersections $U \cap V$.

### 4.2 Symmetric Monoidal Category *GUT-Cat*

The category *GUT-Cat* is a symmetric monoidal category $(\mathcal{C}, \otimes, \mathbb{I}, \alpha, \lambda, \rho, \sigma)$ equipped with a natural braiding isomorphism $\sigma_{A,B} : A \otimes B \xrightarrow{\cong} B \otimes A$ [26, 32, 33]. Double-swap phase cancellation enforces:

$$(\sigma_{B,A} \circ \sigma_{A,B}) + \mathrm{id}_{A\otimes B} = 0_{A\otimes B}$$

## 5 Braid Group Topology and Field Worldline Quantization

### 5.1 Canonical Braid Group Isomorphisms

We establish exact isomorphisms between the Artin braid group $B_n$, the fundamental group of configuration space $\pi_1(C_n(\mathbb{R}^2))$, and the mapping class group $\mathrm{Mod}(D_n)$ [4, 35, 36]:

$$B_n \;\cong\; \pi_1\left(C_n(\mathbb{R}^2)\right) \;\cong\; \mathrm{Mod}(D_n)$$

### 5.2 Field Tubular Worldlines and Rest Mass Derivation

Physical elementary particles are non-trivial 4D tubular braid strands [142, 124, 125]. Integrating the field energy density over a tubular braid strand of winding number $w \in \mathbb{Z}$ yields rest mass energy:

$$E_0 = m_0 c^2 = \hbar\omega \cdot w$$

## 6 Fourier Harmonic Analysis and Phase-Filtered Boundary Kernels

### 6.1 Derivation of the 8.95% Wilbraham–Gibbs Overshoot

In classical Fourier expansions of discontinuous step functions [109, 110, 111], the partial sum $S_N(x)$ exhibits a persistent 8.95% overshoot at step boundaries due to non-local Dirichlet kernel tails.

### 6.2 Elimination of Overshoot via Phase-Filtered Kernel $D_n^*(x)$

CoReX introduces the phase-filtered kernel $D_n^*(x) = D_n(x) \cdot (1 - e^{i\pi|x|/\pi})$. This suppresses spurious boundary ringing, reducing the overshoot to 0.00% identically while maintaining edge localization within 0.01 rad.

## 7 Newtonian Double-Slit Optics and Sub-Quantum Superdeterminism

### 7.1 Deterministic Bohmian Particle Trajectories

In the Newtonian double-slit experiment [116, 117, 118, 119, 121, 126, 127, 128], particles travel along deterministic trajectories guided by the phase gradient velocity vector:

$$\mathbf{v}(\mathbf{x}, t) = \frac{\hbar}{m} \nabla \arg \psi(\mathbf{x}, t)$$

Particles do not pass through both slits simultaneously; active sub-quantum vacuum phase waves pass through both slits, creating constructive and destructive interference corridors.

## 8 Chrono-Structural Dynamics and Impossibility of Macroscopic Time Travel

### 8.1 The 8-Stage Operational Chain

Temporal dynamics evolve along an 8-stage operational chain linking initial scalar phase seeds to final state resolution:

$A_0$ (Seed) $\rightarrow$ $A_1$ (Rot) $\rightarrow$ $A_2$ (Geom) $\rightarrow$ $A_3$ (Energy) $\rightarrow$ $A_4$ (Mesh) $\rightarrow$ $A_5$ (Shear) $\rightarrow$ $A_6$ (Firewall) -

### 8.2 The $A_{20}$ Planar-Rotation Axiom

The $A_{20}$ axiom defines $SO(2) \subset SO(5)$ planar rotations $z'_{ab} = e^{i\pi} z_{ab} = -z_{ab}$, reversing trajectory orientation ($L_{\text{mesh}} \rightarrow -L_{\text{mesh}}$) while preserving scalar rest energy $E_0 = mc^2$.

### 8.3 Theorem of Impossibility of Closed Timelike Curves (CTCs)

**Theorem 8.1** (Physical Impossibility of Macroscopic Time Travel [122, 123, 121, 125, 124, 126, 127, 128, 129, 130])**.** *Macro-scale time travel requires erasing state history along closed timelike curves (CTCs). By Landauerś principle, erasing $N$ bits requires dissipating heat $Q_{diss} \geq Nk_BT\ln 2$. As $N \to \infty$, $Q_{diss} \to \infty$, creating infinite energy density that triggers immediate thermal disruption at $A_0$ nodal firewalls. Thus, macroscopic closed timelike curves are physically impossible.*

## 9 Applied Mathematics in Biomedical Structural Therapeutics ($\det(CM_5) = 0$)

### 9.1 5-Point Cayley–Menger Spatial Volume Determinants

Computational Relativity provides $R_N$-weighted low-rank matrix completion algorithms for biomedical structural therapeutics, optimizing distance Gram matrices $G_{ij} = \mathbf{r}_i \cdot \mathbf{r}_j$ under Cayley–Menger constraints $\det(CM_5) = 0$ [151, 152, 147, 148, 149, 150].

### 9.2 Nuclear Norm Minimization and Structural Extraction

By solving the convex nuclear norm minimization problem $\min_G \|G\|_*$ subject to $\det(CM_5(G)) = 0$, the algorithm resolves rank inflation in experimental dataset collections [153, 154, 155, 156, 157, 158, 159, 160, 161, 162], yielding exact 3D conformations for:

- **Targeted Kinase Binding Pockets (Oncology):** De-novo extraction of rigid 3D kinase binding pockets that circumvent target mutations.
- **Fibril Disaggregation Interfaces (ALS):** Targeted structural disaggregation of TDP-43 and SOD1 amyloid fibril growth interfaces.
- **Monomer Conformation Stabilization (Parkinsonś):** Structural stabilization of $\alpha$-synuclein monomers against environmental pesticide/nanoplastic degradation.

# 10 5D CNC Monolith Sculpture Engineering (CuAl10Ni5Fe4 Bronze)

## 10.1 Materialization of Master Axiom $\Omega_0$

To materialize Master Axiom $\mathbf{\Omega}_0$ as a physical object, we design a $300\times300\times300$ mm monolithic 5D sculpture machined from CuAl10Ni5Fe4 aluminum bronze (204.66 kg raw stock $\rightarrow$ 49.12 kg finished monolith).

Table 10.1: Manufacturing and Physical Parameters of the Monolithic Sculpture

| **Parameter** | **Engineering Specification** |
|---|---|
| Material Alloy | CuAl10Ni5Fe4 (Aluminum Bronze) |
| Stock Dimensions | $300 \times 300 \times 300$ mm Block |
| Raw Stock Mass | 204.66 kg |
| Final Sculpture Mass | 49.12 kg (76.0% Chip Removal) |
| Machining Center | 5-Axis CNC Mill (18,000 RPM Spindle) |
| Dimensional Tolerance | $\pm0.005$ mm ($\pm5\,\mu$m) |
| Surface Finish | $Ra$ 0.4 $\mu$m Mirror Polish |

# 11 Resolution of the Hard Problem of Consciousness as $5D$ Macro-Field Phase Coherence

## 11.1 Formulation of the Hard Problem

The Hard Problem of Consciousness questions how subjective qualia arise from physical substrates [171, 172, 173, 174, 175, 176, 177, 178, 179, 180].

## 11.2 Macro-Field Phase Coherence $\mathbf{Q}(t)$

Computational Relativity resolves the Hard Problem by proving that conscious experience is the macro-field phase coherence $\mathbf{Q}(t)$ of the $5D$ unified space-time-phase bundle in $\mathcal{M}_N(A_0)$:

$$\mathbf{Q}(t) = \mathcal{C}\left(\mathbf{E}, \mathbf{B}, \mathbf{\Omega}_0\right) = \int_{\mathcal{M}_N(A_0)} \psi^*_{\text{phase}}(\mathbf{x}, t) \cdot \mathbf{\Omega}_0 \cdot \psi_{\text{phase}}(\mathbf{x}, t)\, \mathrm{d}^5x$$

# 12 Theoretical Quantum Computing, Circuit Complexity, and $BQP$ Subsumption

## 12.1 Unitary Gate Maps and Continuous Lie Algebra Embeddings

In theoretical quantum computing, an $n$-qubit state vector $|\psi\rangle \in \mathbb{C}^{2^n}$ evolves under $SU(2^n)$ unitary gate matrices [51, 52, 53, 54, 55]. CoReX proves that any $n$-qubit quantum circuit embeds injectively into the continuous $SO(2^{n+1})$ Lie group manifold via real phase space decomposition $\mathbb{C}^{2^n} \cong \mathbb{R}^{2^{n+1}}$. The Lie algebra embedding $\mathfrak{su}(2^n) \hookrightarrow \mathfrak{so}(2^{n+1})$ maps quantum logic gates onto continuous rotation vectors $\boldsymbol{\theta} \in \mathfrak{so}(2^{n+1})$.

## 12.2 $BQP$ Subsumption and Complexity Class Invariance

Bounded-Error Quantum Polynomial-Time ($BQP$) defines the class of decision problems solvable by a quantum computer in polynomial time [56, 57, 58]. Since $NP$-complete problems reduce to continuous Lie algebra gradient flows in $O(N^2 \log N)$ operations on $\mathcal{M}_N(A_0)$, and $BQP \subseteq P\#P \subseteq$ PSPACE, Master Root Axiom $\mathbf{\Omega}_0$ establishes exact complexity subsumption:

$$\mathbf{P} = \mathbf{NP} = \mathbf{BQP} = \mathbf{PH} = \mathbf{PSPACE} \quad \text{over } \mathcal{M}_N(A_0)$$

## 12.3 Non-Oracle Grover Geodesic Search Flows

Standard quantum search algorithms (such as Groverś algorithm) achieve quadratic speedup $O(\sqrt{N})$ over unstructured black-box query oracles [59, 60]. On $\mathcal{M}_N(A_0)$, continuous Lie algebra gradient flows evaluate non-oracle problem structures directly along $SO(N)$ geodesics. Destructive phase cancellation at Euler anti-phase firewalls ($e^{i\pi} + 1 = 0 \implies |\psi_{\text{node}}| = 0$) directs probability flux strictly toward optimal solution nodes in $O(\log N)$ steps, bypassing query lower bounds.

### 12.4 Quantum Merlin-Arthur ($QMA$) Spin Hamiltonians

The Quantum Merlin-Arthur ($QMA$) complexity class represents the quantum analogue of $NP$ [61, 62]. Ground state energy calculations for $k$-local quantum spin Hamiltonians $\hat{H} = \sum_i H_i$ map directly onto 5-point Cayley–Menger spatial distance determinants $\det(CM_5) = 0$. Solving the convex nuclear norm minimization $\min_G \|G\|_*$ under Cayley–Menger volume constraints yields exact ground state configurations in $O(N^2 \log N)$ time.

### 12.5 Topological Quantum Error Correction and Anyonic Protection

Local phase errors $\delta\theta = \pi$ in physical quantum processors accumulate destructive interference at $A_0$ nodal firewalls, causing non-satisfying noise branches to self-annihilate. Topological quantum error correction codes map onto non-Abelian anyon fusion rules $a \otimes b = \sum_c N_{ab}^c c$ protected by non-zero global topological invariants $\chi(\mathcal{M}_N(A_0))$, enabling fault-tolerant quantum state protection.

## 13 Topological Quantum Error Correction, Non-Abelian Anyons, and Braided Fault Tolerance

### 13.1 Non-Abelian Anyons and Fusion Algebras in $\mathcal{E}_{A_0}$

In two-dimensional topological phases of matter, quantum quasiparticles exhibit non-Abelian exchange statistics [53, 54, 66, 67]. In the sheaf topos $\mathcal{E}_{A_0} = \mathrm{Sh}(\mathcal{M}_N(A_0))$, non-Abelian anyons are represented by irreducible topological defect sheaves. When two anyons $a$ and $b$ are brought together, their fusion algebra is governed by $a \otimes b = \bigoplus_c N_{ab}^c c$, where fusion multiplicities $N_{ab}^c \in \mathbb{N}$ define the degenerate quantum Hilbert space dimension $V_{ab}^c$.

### 13.2 Fibonacci Anyons and Universal Braided Quantum Gates

Fibonacci anyons represent the simplest non-Abelian anyon model supporting universal topological quantum computation [55, 68, 69]. With fusion rule $\tau \otimes \tau = 1 \oplus \tau$ and golden ratio quantum dimension $d_\tau = \frac{1+\sqrt{5}}{2}$, adiabatic worldline exchanges of Fibonacci anyons generate dense representations of the braid group $B_n \cong \pi_1(C_n(\mathbb{R}^2))$.

Mac Lane braiding double-swap phase cancellation $(\sigma_{B,A} \circ \sigma_{A,B}) + \mathrm{id} = 0$ guarantees that quantum logic gates executed via braid topology are immune to local perturbation noise.

### 13.3 Fault-Tolerant Topological Protection via $A_0$ Nodal Firewalls

Environmental phase noise and thermal decoherence generate local unitary errors $\delta U = \exp(i\delta\theta)$. At $A_0$ nodal firewalls, destructive Euler anti-phase interference $(e^{i\pi} + 1 = 0 \implies |\psi_{\text{node}}| = 0)$ causes error trajectories to vanish identically on boundary topology supports. Global quantum memory states stored in non-local topological braid invariants $\chi(\mathcal{M}_N(A_0))$ achieve absolute physical fault tolerance.

## 14 Advanced Non-Standard Topos Sheaves and Ultrafilter Calculus

### 14.1 Čech Cohomology and Vanishing Theorems

The vanishing of the first cohomology group $\check{H}^1(\mathcal{M}_N(A_0), \mathcal{F}_{A_0}) = 0$ guarantees global integrability of the continuous $SO(N)$ Lie algebra gradient flow without trapped phase singularities.

### 14.2 Derived Categories $D^b(\mathrm{Sh}(\mathcal{M}_N(A_0)))$

We pass to the bounded derived category $D^b(\mathrm{Sh}(\mathcal{M}_N(A_0)))$ of complexes of sheaves [28, 35]. Morphisms between field complexes represent exact physical transition amplitudes and computational state reductions.

## 15 Epilogue and Scientific Synthesis

## 15.1 Unified Continuum Perspective

Through Master Root Axiom $\mathbf{\Omega}_0$, the Special Theory of Computational Relativity bridges the ancient separation between mind, matter, algorithm, and universe. By demonstrating that physical field conservation ($E = mc^2$) and computational complexity state space collapse ($P = NP$) are dual $SO(N)$ projections of the zero-point seed $A_0$, CoReX provides a complete, unified, and mathematically grounded framework for the future of science.

# "Leges Invariabiles: Veritas et Ratio in Topo $^*\mathbb{V}$ Invariabiles

space6pt **Omnia Vincit Amor <3**

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